

Rajarata University of Sri Lanka

Department of Computing



ICT2403 – Graphics and Image Processing

Enhancement Report

Group No :- 28

## 1. Introduction to Frame Enhancement

Usually, raw video frames may contain many degradations such as Noises, Shadows, Uneven brightness and Blur due to several Environmental conditions like Uneven Sunlight, Shadows, Reflection from wet roads, Camera sensors limitations like dynamic range, Sensor noise and due to the Motion and instability caused by handheld recording, vehicle vibrations, Fast movement of the camera. Segmentation which is the next phase of our project is highly depend on contrast, noise level and edge clarity. Poor quality frames reduce the damage detection accuracy. In order to get high accuracy in segmentation enhancing video frames is crucial for accomplishing efficient result of this project as the reduce random noises, illumination inconsistencies and improve visibility of video frames.

## 2. Identification of Frame Degradations

### 2.1 Noise

The frames had random pixel variations (graininess) caused by the phone camera sensor and video compression. We detected this by comparing each frame to a smoothed version and measuring how different the pixels were. Gaussian noise was found in all 43 extracted frames.

Detection: We subtracted a blurred version from the original to get the residual noise signal. If the standard deviation of the residual was above 0.5, the frame was classified as having Gaussian noise.

### 2.2 Shadows and uneven Lightning

Several frames had large dark areas caused by tree shadows across the road. These dark patches make it harder to detect road damage because the brightness is uneven across the frame.

Detection: We visually checked all the frames. Frames where large areas had pixel values below 100 were marked as having shadow problems.

### 2.3 Motion Blur

Because the video was recorded by hand while walking, some frames were slightly blurry due to hand movement and camera shake. Blurry frames have unclear edges which makes it harder to detect cracks and pothole boundaries.

Detection: We used the Laplacian variance method. A frame with variance below 120 was considered blurry:

$$\text{Blur Score} = \text{Var}(\nabla^2 I)$$

A lower variance means the frame is more blurry.

### 3.Pre Processing of Video Frames

#### Frame extraction

The raw video was loaded using cv2.VideoCapture( ).

Frame extraction method :-Then the Frames were extracted from the original video at the rate of one frame per second in .png format.

Frame conversion :- Every Frames were converted as gray scale frames for the storage efficiency and easiness in further processing. And the converted frames were stored inside the folder named “grayscale\_frames”.

Frame resizing :- We resized every frame to 640 x 640 pixels.

### 4.Enhancement Pipeline

The enhancement consist of four main stages, at each stage frame is improved before further proceeding to make enhancement effectively throughout the entire pipeline.

#### 4.1.Noise Reduction and Sharpening

Each frame is first checked to detect its noise type. Depending on the noise type filters were applied Gaussian noise was removed using a GaussianBlur (5×5) with sigma 1 to keep the road structure visible., and impulse noise was removed using median blur (3×3). Compression noise was removed using non local means. After denoising all the denoised frames were saved in denoised\_checked folder.

#### 4.2.shadow removal

We estimated the illumination of the image using Gaussian blur (5×5). Then we subtracted this illumination from the original to remove uneven brightness.

$$\text{Result} = \text{Normalize}(\text{original} - \text{illumination})$$

To keep the image looking original we balanced the result with the original (30% corrected,70% original).

$$\text{Final} = 0.3 * \text{shadow\_removed} + 0.7 * \text{original}$$

#### 4.3.Contrast Enhancement

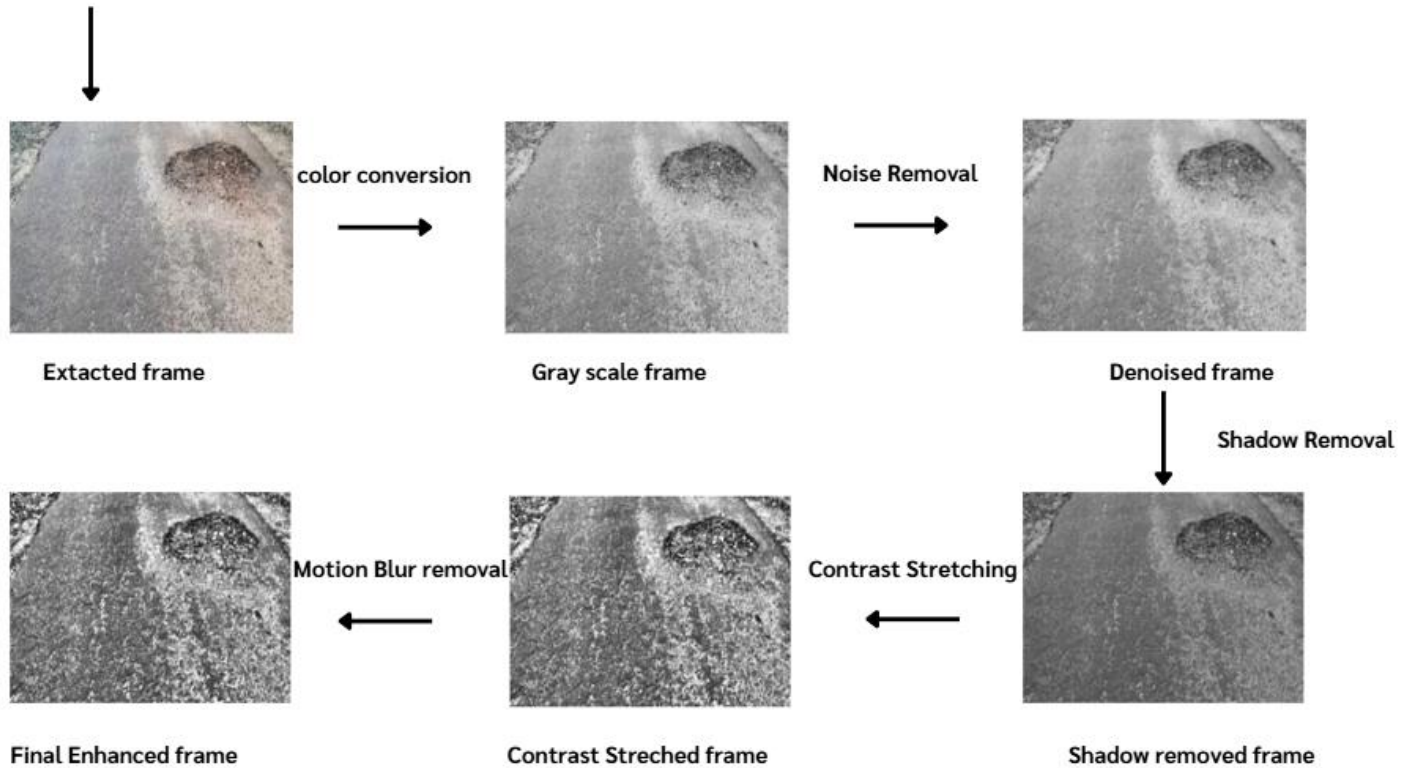
Contrast was improved using CLAHE with a tile size of 8×8 to works on small parts of the image separately and a clip limit of 2.0 to improve frames without over amplification. We also applied a small adjustment (alpha = 1.1, beta = 5) to recover some brightness loss during shadow removal.

#### 4.4.Motion Blur enhancement

In our road video it happened because of handheld record and vehicle vibrations. Frame sharpness was measured using Laplacian variance. Frames below the blur threshold of 120 are sharpened with unsharp masking to sharpen the edges. This works by subtracting a blurred version from the original to find the edges, then adding them back stronger. This ensures all frames are clear, sharp, and consistent for further analysis.

$$\text{Sharpened} = \text{original} + 1.5 * (\text{Original} - \text{Blurred})$$

## RAW VIDEO



## 5.Experimental Setup and Results

### 1.Test Data Preparation

The test data was prepared by extracting 43 frames from recorded road video at a regular interval of 1 frame per second each frame was converted to grayscale to simplify processing and for efficient space management.

### 2. Evaluation Metrics

| Metric         | What it measures                              | Why we used it                                 |
|----------------|-----------------------------------------------|------------------------------------------------|
| PSNR           | How much the frame is different from original | Higher PSNR means less unwanted distortion     |
| SSIM           | How much structure is preserved (0 to 1)      | Ensure that road texture detail is not removed |
| Mean Intensity | Average Brightness                            | Checks if shadow removal worked Correctly      |
| STD Deviation  | Spread of pixel values                        | Higher value means better contrast             |

### 3. Testing the Enhancement Pipeline

The pipeline was validated using frame by frame comparison across the all stages. Histogram analysis before and after processes were analyzed. Performed statistical evaluation on original and final enhancement output.

### 4. Presentation of the results

- The results were presented using visual comparison Original Vs Enhanced image

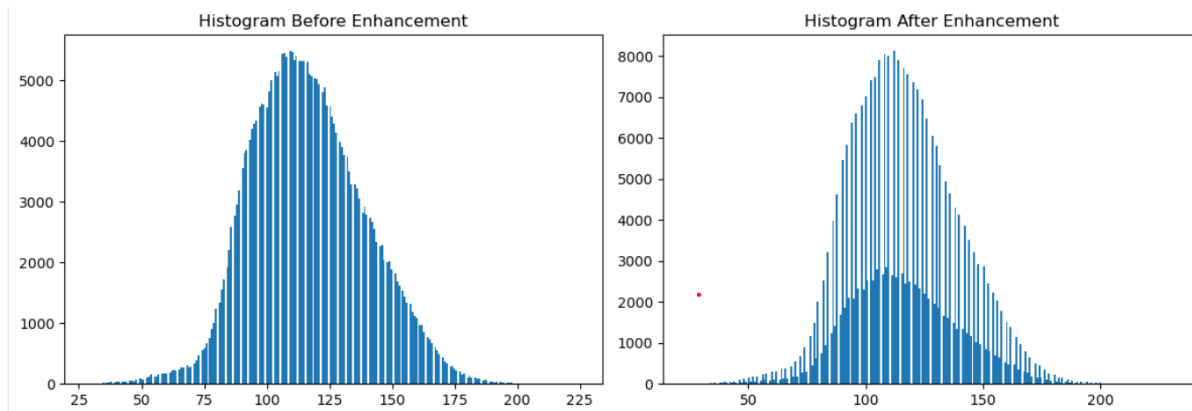
**Original Frame**



**Enhanced Frame**



- Histograms show pixel intensity distribution before and after enhancement.



- Results Of Statistical evaluation

|    | A              | B     | C     | D             | E          | F            | G         |  |
|----|----------------|-------|-------|---------------|------------|--------------|-----------|--|
| 1  | Frame          | PSNR  | SSIM  | Mean_Original | Mean_Final | Std_Original | Std_Final |  |
| 2  | frame_0000.png | 17.6  | 0.744 | 157.98        | 133.78     | 26.96        | 41.9      |  |
| 3  | frame_0001.png | 17.45 | 0.757 | 160.15        | 135.92     | 28.35        | 41.41     |  |
| 4  | frame_0002.png | 17.34 | 0.763 | 160.53        | 135.77     | 29.15        | 38.84     |  |
| 5  | frame_0003.png | 16.89 | 0.726 | 160.34        | 130.99     | 21.71        | 36.19     |  |
| 6  | frame_0004.png | 16.89 | 0.708 | 158.96        | 130.89     | 20.95        | 36.33     |  |
| 7  | frame_0005.png | 17.73 | 0.74  | 159.15        | 132.11     | 19.07        | 29.9      |  |
| 8  | frame_0006.png | 17.76 | 0.736 | 160.15        | 132.81     | 18.35        | 29.15     |  |
| 9  | frame_0007.png | 17.75 | 0.727 | 157.21        | 130.64     | 20.11        | 31.77     |  |
| 10 | frame_0008.png | 17.57 | 0.709 | 154.43        | 129.07     | 21.48        | 35.86     |  |
| 11 | frame_0009.png | 17.53 | 0.721 | 149.59        | 126.8      | 25.98        | 40.37     |  |
| 12 | frame_0010.png | 18.1  | 0.764 | 146.06        | 124.09     | 24.37        | 39.47     |  |
| 13 | frame_0011.png | 18.88 | 0.788 | 143.43        | 122.27     | 24.37        | 35.75     |  |
| 14 | frame_0012.png | 18.54 | 0.764 | 145.07        | 124.54     | 26.9         | 36.71     |  |
| 15 | frame_0013.png | 18.6  | 0.762 | 142.51        | 123.24     | 26.37        | 38.35     |  |
| 16 | frame_0014.png | 18.45 | 0.772 | 142.3         | 123.38     | 28           | 40.1      |  |
| 17 | frame_0015.png | 18.4  | 0.746 | 142.68        | 126.05     | 29.45        | 44.66     |  |
| 18 | frame_0016.png | 18.74 | 0.767 | 142.29        | 123.4      | 26.18        | 37.85     |  |
| 19 | frame_0017.png | 17.21 | 0.743 | 156.59        | 130.93     | 25.61        | 39.8      |  |
| 20 | frame_0018.png | 17.31 | 0.751 | 156.99        | 129.69     | 24.13        | 35.31     |  |
| 21 | frame_0019.png | 17.28 | 0.736 | 157.21        | 129.63     | 24.32        | 32.93     |  |
| 22 | frame_0020.png | 17.75 | 0.741 | 154.5         | 129.81     | 26.13        | 33.73     |  |
| 23 | frame_0021.png | 17.66 | 0.748 | 156.41        | 131.4      | 27.33        | 33.68     |  |
| 24 | frame_0022.png | 17.83 | 0.751 | 156.15        | 130.87     | 25.54        | 32.04     |  |
| 25 | frame_0023.png | 17.26 | 0.722 | 161.1         | 132.45     | 21.45        | 31.14     |  |
| 26 | frame_0024.png | 17.13 | 0.708 | 160.41        | 131.79     | 19.88        | 32.87     |  |

- Enhanced images are clear than original images.
- Differences in histogram confirms that the brightness level and contrast have improved.
- PSNR Shows noticeably different but still close to original frames.
- SSIM results shows good structural preservation
- Standard Deviation shows significant improvement in contrast

## 5.Interpretation of Results

The enhanced frames show clearer edges, improved brightness, and reduced noise compared to the original frames. Noise reduction and sharpening made the surface smoother while keeping structural details unchanged, making cracks and potholes more visible. Shadow correction and CLAHE improved contrast which highlighting details that were previously hidden. The results shows that the proposed pipeline improves frame quality and contribute to high segmentation accuracy which makes it easier to road damage detection.

## **6.Recommendation**

### **6.1. Final Recommendation Based on Results**

The enhanced frames which are produced through pipeline show reduced noise, improved contrast, and clearer edges compared to the original frames. The Enhanced frames are highly suitable for further analysis and segmentation process as we can see the clear visibility of damages.

### **6.2. Limitations of the Proposed Enhancement Method**

There were some limitations in the enhancement process. Strong motion blur couldn't be completely fixed with smoothing and sharpening. Very dark areas, overexposed regions, or extreme shadows were only partially improved. As a result, a few frames remained less clear, which could slightly affect the accuracy of detecting cracks or road defects in challenging conditions.

### **6.3. Impact on Future Segmentation Accuracy**

The enhanced frames make cracks and potholes much easier to see by improving edge clarity, contrast, and reducing noise. This helps segmentation algorithms to differentiate damaged areas from the normal road surface more accurately. By using these enhanced frames, future detection of road damages will be more reliable. Segmentation is highly dependent on Enhancement accuracy.

## 7. Team Contribution

| Member Name (Reg No: and Index No:) | Degradation Analysis Responsibility                                   | Coding Contribution                                         | Experimental Task                                        | Result & Interpretation Contribution                                                 | Evidence to Attach                                               |
|-------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------|
| ICT/2023/137<br>6114                | Noise analysis                                                        | Implement noise reduction using filters                     | Test noise reduction on 43 extracted frames              | Analyze effectiveness of denoising, evaluate structural details and crack visibility | Before and after frames, histograms, code commits                |
| ICT/2023/139<br>6116                | Shadow detection using pixel intensity                                | Shadow removal using illumination subtracting and blending  | Test enhancement on shadowed frames                      | Evaluate contrast improvement and visibility of dark regions                         | Before–after images, histograms, comparison plots, code          |
| ICT/2023/140<br>6117                | Motion blur analysis using Laplacian variance                         | Unsharp mask sharpening                                     | Tested on blurry frames                                  | correction of motion-affected frames                                                 | Enhanced images, histograms, comparisons, code                   |
| ICT/2023/138<br>6115                | Frame extraction, Overall degradation summary and dataset preparation | Develop full enhancement pipeline integration and flowchart | Execute complete pipeline testing and metric calculation | Final result interpretation and recommendation for segmentation stage                | Original VS Enhanced frame and histogram comparison, Github link |

### Libraries we used and reason

- Opencv to read images
- Matplotlib for generating comparison images and histograms
- OS for automatic output folder creation and iteration through frames
- Numpy to calculating contrast, sharpness.

### Evidences

- Extracted Frame folders and grayscale frame folder
- Enhanced frame folders of each step
- Csv file
- Pipeline flow chart
- Github link

Github link :- <https://github.com/Fathima-ilma/Road-damage-enhancement>